HealthTrack: Predicting Hospital Readmission

Problem Statement

Mini Project – Data Science | Dept. of Computer Science | 2024–25

Background

Unplanned hospital readmissions within 30 days of discharge represent a major challenge for hospitals worldwide. They are widely used as a proxy measure of the quality of inpatient care and care transition. Studies indicate that approximately 15–20% of patients are readmitted within 30 days, and a significant proportion of these readmissions are considered preventable with proper intervention.

Problem Definition

Given the clinical data of a patient at the time of discharge, predict whether that patient will be readmitted to the hospital within 30 days. This is a binary classification problem where:

Class 1 (Positive): Patient is readmitted within 30 days

Class 0 (Negative): Patient is not readmitted within 30 days

Objectives

- Build a supervised machine learning model to predict 30-day readmission.
- Identify the most important clinical features that contribute to readmission risk.
- Classify patients into Low, Medium, or High risk tiers at discharge.
- Develop an accessible web interface for use by non-technical healthcare staff.
- Evaluate the model using accuracy, ROC-AUC, precision, recall, and F1-score.

Scope and Constraints

The system operates on structured tabular patient data collected at the time of discharge. The current implementation uses synthetic data for demonstration and academic purposes. In a production environment, the model would need to be retrained on real Electronic Health Record (EHR) data and validated against clinical outcomes. The system does not replace clinical judgment and is intended as a decision-support tool only.

Expected Outcome

A trained and evaluated Random Forest model integrated into a Streamlit web application that enables real-time prediction of 30-day readmission risk, along with visual explanations of the model's behavior and dataset characteristics.